

Chapter 1

Theorem Statement

Theorem 1.1 (BC1). (X, d) is a complete metric space. $\{U_i\}_{i \in \mathbb{N}}$ is a collection of open dense sets. $U = \bigcap_{i=1}^{\infty} U_i$ is a dense set.

Proof: $\forall x_0 \in X$ and $\forall r_0 > 0$. We want to show that $B_{x_0}(r_0) \cap U \neq \emptyset$. Note because U_1 is a dense set, $U_1 \cap B_{x_0}(r_0) \neq \emptyset$ further more because U_1 is open there exists $0 < r_1 < \frac{r_0}{2}$ and $x_1 \in U_1 \cap B_{x_0}(r_0)$ where

$$\tilde{B}_{x_1}(r_1) \subset B_{x_1}(2r_1) \subset U_1 \cap B_{x_0}(r_0)$$

Define the sequence (x_i) and (r_i) recursively. Because $U_{n+1} \cap B_{x_n}(r_n)$ is open and nonempty (density of U_{n+1}), $\exists x_{n+1} \in U_{n+1} \cap B_{x_n}(r_n)$ and $r_{n+1} \in (0, \frac{r_n}{2})$ where

$$\tilde{B}_{x_{n+1}}(r_{n+1}) \subset B_{x_{n+1}}(2r_{n+1}) \subset U_{n+1} \cap B_{x_n}(r_n)$$

The sequence x_i is Cauchy. This is because for $m \geq n \geq N$:

$$d(x_m, x_n) \leq r_n \leq r_N < r_0 2^{-N}.$$

$\forall \varepsilon > 0$, let $N = \left\lceil -\log_2 \left(\frac{\varepsilon}{r_0} \right) \right\rceil$, thus $m \geq n \geq N$:

$$d(x_m, x_n) \leq r_0 2^{-N} < \varepsilon.$$

$r_n \rightarrow 0$ geometrically. By completeness of X , $\exists y \in X$ such that $x_i \rightarrow y$. For all n , $x_k \in \tilde{B}_{x_n}(r_n)$ for $k \geq n$. Since closed sets contain all limit points, $x^* \in \tilde{B}_{x_n}(r_n) \subset U_n$ for all n . thus $x^* \in U \cap B_{x_0}(r_0)$. $\square$

Theorem 1.2 (BC2). (X, d) is a complete metric space. $\{V_i\}_{i \in I}$ be a collection closed sets with no interior.

$$X \neq \bigcup_{i \in I} V_i$$

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References